

GENERAL INSTRUCTIONS: Section A (1 mark each), Section B (2 marks each), Section C (3 marks each), Section D (4 marks each)

SECTION - A : OBJECTIVE & VERY SHORT QUESTIONS

(1 Mark Each)

1. The number 3 is: [1 Mark]
(A) An even number
(B) A composite number
(C) A prime number
(D) None of these
2. Zeroes of the quadratic polynomial $x^2 - 2x - 8$ are: [1 Mark]
(A) 2, -4
(B) 2, 4
(C) -2, -4
(D) None of these
3. The 8th term of an A.P. 10, 7, 4, ... is: [1 Mark]
(A) -11
(B) -10
(C) -8
(D) None of these
4. Distance between the points (4, 3) and (0, 0) is: [1 Mark]
(A) $\sqrt{7}$
(B) 5
(C) 12
(D) None of these
5. If $6x + 3y = C - 3$ and $12x + Cy = C$ have infinitely many solutions, then C is equal to: [1 Mark]
(A) 6
(B) 5
(C) 4
(D) None of these
6. Circumference of a semi-circle with radius 'r' is: [1 Mark]
(A) $2 * \pi * r$
(B) $\pi * r$
(C) $\pi * r^2$
(D) None of these
7. How many parallel tangents can a circle have at the most? [1 Mark]
(A) 2
(B) 1
(C) Infinite
(D) None of these

8. A card is drawn from a pack of 52 cards. What is the probability of getting a heart? [1 Mark]
 (A) $1/26$
 (B) $1/2$
 (C) $1/4$
 (D) None of these
9. $1 - \cos^2 30^\circ$ is equal to: [1 Mark]
 (A) $-\sin^2 30^\circ$
 (B) $\sin^2 30^\circ$
 (C) $-\sin^2 60^\circ$
 (D) None of these
10. The product of roots of the quadratic equation $\sqrt{5}x^2 + 3x - 5 = 0$ is: [1 Mark]
 (A) $\sqrt{5}$
 (B) $1/\sqrt{5}$
 (C) $-\sqrt{5}$
 (D) None of these
11. H.C.F. of any two prime numbers is [1 Mark]
12. The sum of first n natural numbers is $n(n+1)/2$. (State True or False) [1 Mark]
- OR ---
- The next term of the A.P. $\sqrt{27}, \sqrt{48}, \sqrt{75}, \dots$ is $\sqrt{108}$. (State True or False)
13. The value of $\cos \theta$ increases as θ increases from 0° to 90° . (State True or False) [1 Mark]
14. All triangles are similar. (isosceles / equilateral) [1 Mark]
15. The common point of a tangent to a circle and the circle is called the [1 Mark]
16. Define Distance Formula in coordinate geometry. [1 Mark]
- OR ---
- Define the Ordinate of a point.
17. State Basic Proportionality Theorem (Thales' Theorem). [1 Mark]
18. Define Angle of Elevation. [1 Mark]
19. What is the probability of a sure event? [1 Mark]
20. Write the mathematical formula for the total surface area of a solid hemisphere. [1 Mark]

SECTION - B : SHORT ANSWER QUESTIONS

(2 Marks Each)

21. Find the H.C.F. of 510 and 92 using prime factorisation. [2 Marks]
22. A drinking glass is in the shape of a frustum of a cone of height 14 cm. The diameters of its two circular ends are 4 cm and 2 cm. Find the capacity of the glass. [2 Marks]
23. Solve the pair of linear equations: $x + y = 5$ and $2x - 3y = 4$ by elimination method. [2 Marks]
24. Given $15 \cot A = 8$, find $\sin A$ and $\sec A$. [2 Marks]

--- OR ---

Evaluate: $2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$

25. A bag contains 3 red balls and 5 black balls. A ball is drawn at random from the bag. What is the probability that the ball drawn is: (i) red ? (ii) not red ? [2 Marks]
26. A class teacher has the following absentee record of 40 students of a class for the whole term. Find the mean number of days a student was absent: [2 Marks]

Number of Days	Number of Students
0 - 6	11
6 - 10	10
10 - 14	7
14 - 20	4
20 - 28	4
28 - 38	3
38 - 40	1
Total	40

SECTION - C : SHORT ANSWER QUESTIONS

(3 Marks Each)

27. Obtain all other zeroes of $3x^4 + 6x^3 - 2x^2 - 10x - 5$, if two of its zeroes are $\sqrt{5/3}$ and $-\sqrt{5/3}$. [3 Marks]
- Find a quadratic polynomial, the sum and product of whose zeroes are $\sqrt{2}$ and $1/3$, respectively.
28. A fraction becomes $1/3$ when 1 is subtracted from the numerator and it becomes $1/4$ when 8 is added to its denominator. Find the fraction. [3 Marks]
29. Find the roots of the quadratic equation $2x^2 + x - 6 = 0$ by factorisation. [3 Marks]
30. Find the 31st term of an A.P. whose 11th term is 38 and the 16th term is 73. [3 Marks]

--- OR ---

Find the sum of the first 15 multiples of 8.

31. Write all the other trigonometric ratios of angle A in terms of $\sec A$. [3 Marks]
32. Prove that the lengths of tangents drawn from an external point to a circle are equal. [3 Marks]

--- OR ---

The length of a tangent from a point A at distance 5 cm from the centre of the circle is 4 cm. Find the radius of the circle.

33. Find the area of a quadrant of a circle whose circumference is 22 cm. [3 Marks]
34. How many silver coins, 1.75 cm in diameter and of thickness 2 mm, must be melted to form a cuboid of dimensions 5.5 cm x 10 cm x 3.5 cm? [3 Marks]

SECTION - D : LONG ANSWER QUESTIONS

(4 Marks Each)

35. A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the same journey. Find the original speed of the train. [4 Marks]

--- OR ---

Find the roots of $4x^2 + 3x + 5 = 0$ by the method of completing the square.

36. The angles of elevation of the top of a tower from two points at a distance of 4 m and 9 m from the base of the tower and in the same straight line with it are complementary. Prove that the height of the tower is 6 m. [4 Marks]
37. Find the area of the quadrilateral whose vertices taken in order, are (-4, -2), (-3, -5), (3, -2) and (2, 3). [4 Marks]

--- OR ---

If A and B are (-2, -2) and (2, -4), respectively, find the coordinates of P such that $AP = \frac{3}{7} AB$ and P lies on the line segment AB.

38. Prove that the line joining the mid-points of any two sides of a triangle is parallel to the third side. [4 Marks]

--- OR ---

In an equilateral triangle, prove that three times the square of one side is equal to four times the square of one of its altitudes.

39. Draw a pair of tangents to a circle of radius 5 cm which are inclined to each other at an angle of 60 deg. [4 Marks]
40. If the median of the distribution given below is 28.5, find the missing frequencies x and y (Total Frequency = 60): [4 Marks]

Class Interval	Frequency
0 - 10	5
10 - 20	x
20 - 30	20
30 - 40	15
40 - 50	y
50 - 60	5
Total	60

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